Part: **3.3 – METAL ASSEMBLIES**

Chapter: **3345**

Title: **BULKHEADS**

Revision: **0.1**

Date: **19/04/2021**

1 INTRODUCTION

1.1 OBJECT & SCOPE

This chapter intends to present some general rules and recommendations for performing the structural analysis of fuselage bulkheads made of metallic alloys. These elements divide the fuselage in slices, providing support to skin and longitudinal stringer to prevent buckling stability failures, distributing local loads introductions within the airframe (eg. at wing or landing gear junctions) and complete the vessel of pressurized cabins.

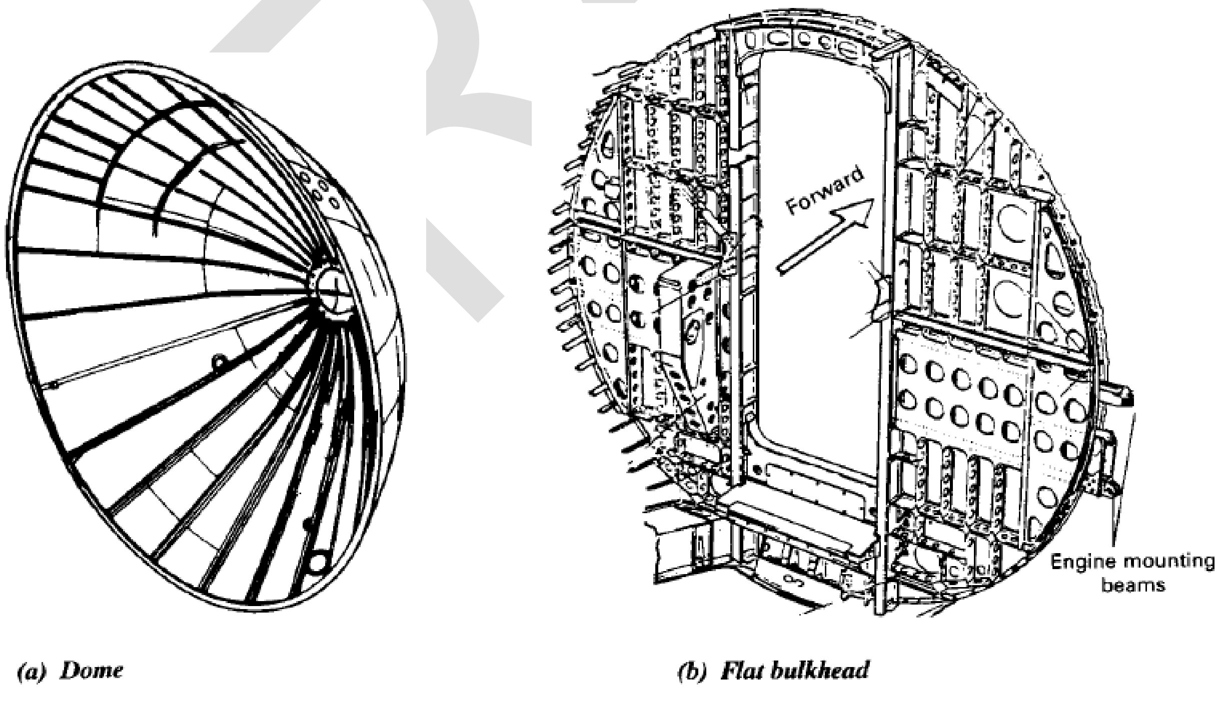


Figure 1. Typical pressure bulkheads (Niu's book Ref. 1)

Some of abovementioned functions are common to ring frames treated in chapter 3340 “FRAMES”, except obviously serving as terminal wall for pressure containment of pressurized cabin.

Fuselage bulkheads can be flat and curved (spherical shells), shall typically include stiffening members on to the shell panel and, in particular in non-pressurized cases, holes of different sizes and geometries. The variety of potential configurations is huge and thus quite difficult to describe analysis processes sufficiently general and exhaustive at the same time.

2 STRUCTURAL ANALYSIS OF BULKHEADS

GFEM vs DFEM

GFEM refinement at bulkhead for detailed stress/strength analysis

Pressure loads when applies

Strength :

- Von-Mises read in FEM vs Ftu (chapter 3102)
- Holes and strength… FEM refined or KT’s application?

Buckling by analysis (SMM – Vol. 3) or directly taken from DFEM if mesh sufficiently refined

- Global stability: FEM
- Flanges local-buckling/crippling: chapter 3130
- Web local buckling: depending FEM mesh refinement and geometries of web bays enclosed among stiffening/support members analytical approaches could be feasible (chapter 3150)

Fastened joint to skin:

- Fastener loads: from FEM shell elements of bulkhead web adjacent to skin
- Bolt, shear-bearing, tension-angle/fitting in chapters 3220, 3230, 3240

The skin shall be considered in this analysis mainly to account its contribution in the stress distribution on the bulkhead. It is not intended in this method to present complete static margins of the fuselage skin.

The strength and stability analysis of the skin shall be fully described in the stiffened skin method described in chapter §3320 “CURVED STIFFENED PANEL” .

3 SIMPLIFIED METHODOLOGIES FOR PRE-SIZING

For a few typical configurations some simplified methodologies are proposed. These methods must be considered coarse, approximated, and should be used only for preliminary studies and presizing activities. Even more, it is strongly recommended, until the characteristic deviation of these methodologies is clearly assessed, to apply generous margin policies to resulting predictions.

3.1 BULKHEADS AT WING JUNCTION STATIONS

The bulkhead-frame type considered herein is flat, holed (of significant sizes, as bays for engines or other systems), not-pressurized, and it is sized by a discrete load introduction at its Y^+ and Y^- ends introduced by the wing junction, when aircraft is subjected to a symmetric load case.

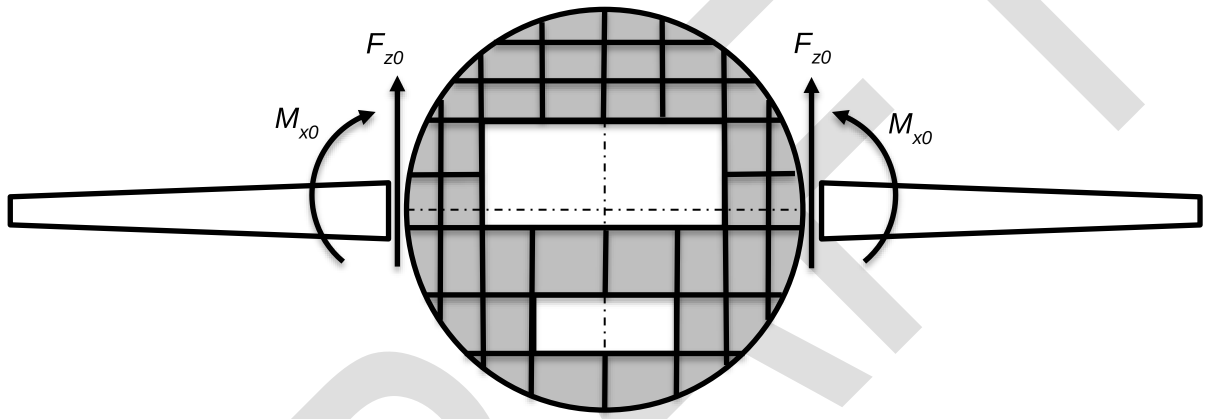


Figure 2. Scheme of bulkhead-frame considered

3.1.1 Internal loads in frame

Main assumptions:

- ✓ The moment M_x introduced by wing loads has to be sustained in bending by the frame and its nearby skin (effective width)
- ✓ The vertical force F_z must be transferred to the fuselage skin through its (fastened) attachment to the frame, along its contour
- ✓ The stress distribution in fuselage skin induced by abovementioned vertical force corresponds to the shear distribution obtained in a beam section with a hollow circle shape of very thin wall/thickness

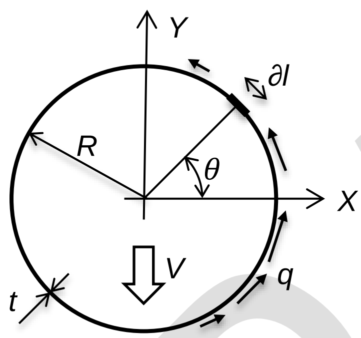
Shear distribution in a thin walled circular hollow section (fuselage skin)

Shear stresses in this type of thin-walled section shall be calculated thanks to the classical theory of elasticity and Collignon's theorem (shear stresses are uniform across the thickness). Using notation in Niu's book (Ref. 2, §6.5), the shear load flow q in the skin shall be:

$$q = \tau \cdot t = V \cdot Q / I$$

where:

V	total shear force sustained by the section
Q	first moment of inertia at a section point
I	bending moment of inertia of the section
τ	shear stress in the skin wall
t	skin thickness



Properties of thin walled circular hollow section:

Section area	$A = 2 \cdot \pi \cdot R \cdot t$
Section inertia	$I = \pi \cdot R^3 \cdot t$

Figure 3. Shear distribution in fuselage skin

Then…

✓ The 1st moment of area for the thin hollow circle section at any point is:

$$Q = \int y\, dA = \int R \sin(\theta)\, t\, R\, d\theta = R^2\, t \int \sin(\theta)\, d\theta = -R^2\, t\, [\cos(\theta)]_{\theta_0}^{\theta}$$

If the origin for the integral is set at $= -\pi/2$, where the shear stress must be 0, it becomes:

$$Q = -R^2\, t \cos(\theta)$$

✓ And the shear load

$$q = \tau \cdot t = V \cdot Q/I = V \frac{-R^2\, t\, \cos(\theta)}{\pi\, R^3\, t} = -\frac{V \cos(\theta)}{\pi\, R}$$

If expressed using the notation initially proposed for the forces in this case, instead of Niu's:

$$q = \tau \cdot t = \frac{2\, F_{z0} \cos(\theta)}{\pi\, R}$$

It can be easily check that this later expression become zero at most upper and lower points of the section ($\theta = \pm\pi/2$) , and that at ($\theta = 0$) the ratio between the calculated load flow q and the average value ($q_{avg} = F_{z0} / [\pi \cdot R]$) results equal to 2, what matches the relation between max. shear stress and the averaged value that can be found in literature for this type of section.

From above expression the shear load that has to be sustained by the fastened attachment between frame and skin can be easily evaluated:

$$q_{max} = 2\, F_{z0} / (\pi\, R)$$

Shear load in frame sections

From "external" applied forces (introduced in frame at the wing junction) and previously calculated shear load distribution reacted on the skin, it shall be possible to evaluate the internal loads (vertical shear load and in-plane bending moment) at any section of the frame, if treated as a beam.

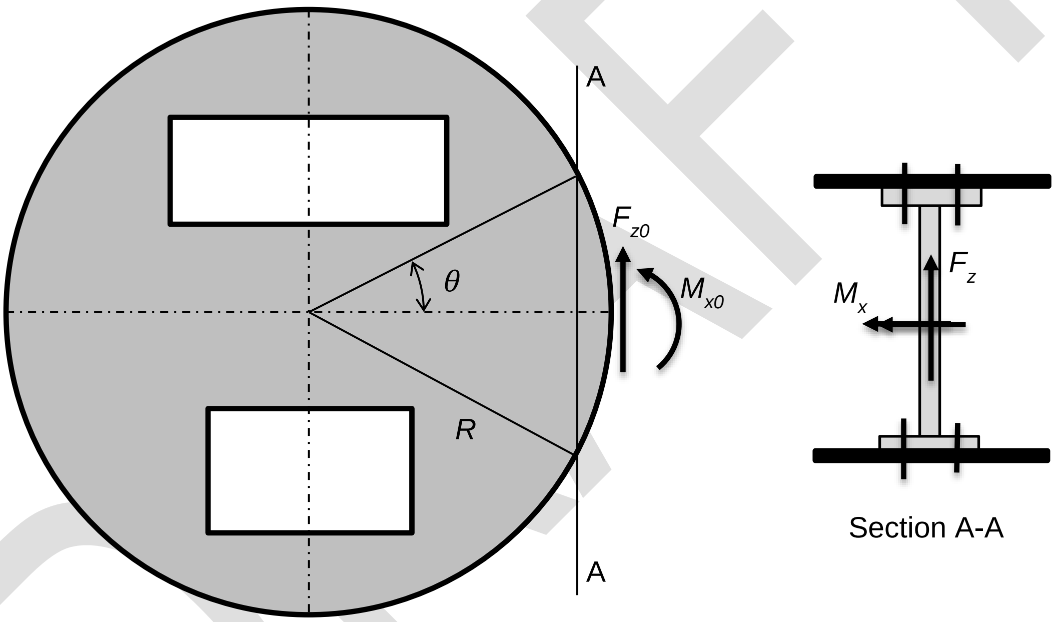


Figure 4. Location of frame section for analysis

The load reacted at fuselage skin can be discounted from the total shear load that has to be sustained by the frame itself:

$$F_z = F_{z0} - \int q\, cos(\vartheta)\, dl = F_{z0} - \int \frac{2\, F_{z0} \cos(\vartheta)}{\pi\, R}\, cos(\vartheta)\, R\, d\vartheta = F_{z0}\left[1 - \frac{2}{\pi}\int cos^2(\vartheta)\, d\vartheta\right]$$

As the integral of the squared cosine function is $\int cos^2(x)dx = x/2 + sin(2x)/4 + c$, then:

$$F_z = F_{z0}\left\{1 - \frac{2}{\pi}\left[\frac{\vartheta}{2} + \frac{\sin(2\vartheta)}{4}\right]_{-\theta}^{+\theta}\right\} = F_{z0}\left\{1 - \frac{1}{\pi}[2\theta + \sin(2\theta)]\right\}$$

It can be easily check that this later expression become zero at section centre ($\theta = \pi/2$), where the forces introduced at wing junctions have been completely transferred to the fuselage skin.

Bending moment in frame sections

A similar exercise can be done for the bending moment sustained at each frame section:

$$M_x = M_{x0} + F_{z0}\,R\,[1-\cos(\theta)] - \int R\,[\cos(\vartheta - \cos(\theta)]\,q\,dl\,\cos(\vartheta) =$$

$$= M_{x0} + F_{z0}\,R\,[1-\cos(\theta)] - \int R\,[\cos(\vartheta - \cos(\theta)]\,\frac{2\,F_{z0}\cos(\vartheta)}{\pi\,R}\,R\,d\theta\,\cos(\vartheta) =$$

$$= M_{x0} + F_{z0}\,R\,[1-\cos(\theta)] - \frac{2\,F_{z0}R}{\pi}\int [\cos(\vartheta - \cos(\theta)]\cos^2(\vartheta)\;d\theta =$$

$$= M_{x0} + F_{z0}\,R\,[1-\cos(\theta)] - \frac{2\,F_{z0}R}{\pi}\left\{\int \cos^3(\vartheta)\;d\theta - \cos(\theta)\int \cos^2(\vartheta)\;d\theta\right\}$$

As the integral of the cubic cosine function is …

$$\int cos^3(x)dx = sin(x) - sin^3(x)/3 + c$$

Then:

$$M_x = M_{x0} + F_{z0}\,R\,[1-\cos(\theta)] - \frac{2\,F_{z0}R}{\pi}\left\{\left[\sin(\vartheta) - \frac{\sin^3(\vartheta)}{3}\right]_{-\theta}^{+\theta} - \cos(\theta)\left[\frac{\vartheta}{2} + \frac{\sin(2\vartheta)}{4}\right]_{-\theta}^{+\theta}\right\} =$$

$$= M_{x0} + F_{z0}\,R\,[1-\cos(\theta)] - \frac{2\,F_{z0}R}{\pi}\left\{\left[2\sin(\theta) - \frac{2\sin^3(\theta)}{3}\right] - \cos(\theta)\left[\theta + \frac{\sin(2\theta)}{2}\right]\right\}$$

The last term of this expression become zero at section ($\theta = 0$), and, at centre section ($\theta = \pi/2$) :

$$M_x = M_{x0} + F_{z0}\,R\left(1 - \frac{8}{3\,\pi}\right)$$

Simplified expressions for shear and bending loads in frame sections

In first approximation, use at any frame section:

$F_z = F_{z0}$ (*)

$M_x = M_{x0} + 0.15 \cdot R \cdot F_{z0}$ (**)

(*) This conservative at centre sections but not expected to be a sizing condition

(**) Maximum reached at centre section, normally not too far from the value of M_{x0} at frame end points

3.1.2 Stress distribution at frame sections

Stress within the frame section shall calculated using classical straight beams theory (which is obviously not realistic for sections of a fuselage frame and will probably introduce a significant error levels in calculated stresses).

Section adjacent to wing junction

Stress analysis:

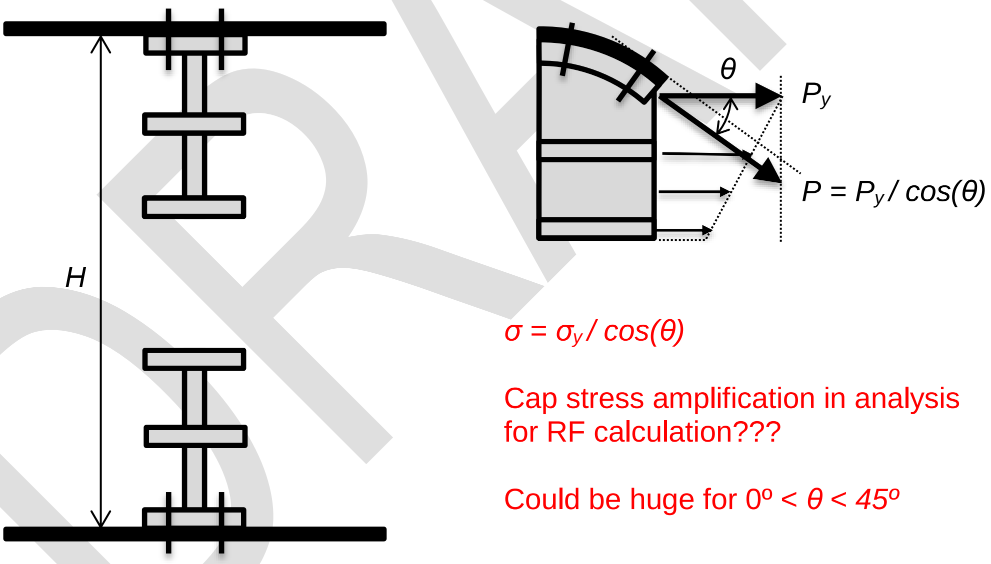
- ✓ Mean shear stress: $\tau = F_{z0} / (h \cdot t)$ …with *h* the effective height
- ✓ Max. bending stress: $\sigma = 6 \cdot M_{x0} / (t \cdot h^2)$

Strength:

- ✓ Reserve Factor: $RF = 1 / [(\sigma/F_{tu})^2 + (\tau/F_{su})^2]^{0.5}$

Intermediate sections

The frame section shall normally include horizontal stiffening members at different heights, as shown in following figure.



$\sigma = \sigma_y / cos(\theta)$

Cap stress amplification in analysis for RF calculation???

Could be huge for $0^o < \theta < 45^o$

Figure 5. Typical frame section

Stress analysis:

- ✓ Mean shear stress: $\tau = F_z / (A_{web})$ …with A_{web} the net section area of the web
- ✓ Max. bending stress: $\sigma_y = M_x (z - z_{cg}) / I$ …with *I* the bending moment of inertia (*)

(*) The bending moment of inertia of the section shall be calculated adding the contribution of a full effective skin width equal to the frames pitch. This is implicitly assuming that skin has to be sizing stable (no buckling allowed) up to ultimate loads. If skin is made of a material different from that of the frame/bulkhead, its corrected thickness for calculating the inertia moment shall be:

$t_{skeq} = t_{sk} \cdot E_{sk} / E_{fr}$ … or use the alternative formulation $\varepsilon_y = M_x (z - z_{cg}) / EI$

Strength:

- ✓ Frame: $RF = 1 / [(\sigma/F_{tu})^2 + (\tau/F_{su})^2]^{0.5}$
- ✓ Skin: Perform the strength check as applicable for the type of material (metal or composite), imposing strain compatibility to get the right internal loads before calculating the static margins

Bucking stability:

Using above discussed stress distribution, apply validated methods to check…

- ✓ in the frame/bulkhead: the local stability and/or crippling of caps and stiffeners (in compression) and web subpanels (in compression and shear)
- ✓ in the skin: its local stability considered simply supported between 2 consecutive frames

3.1.3 Further notes

- ✓ For non-circular frames with width greater than height (eg. in twin engine fighters), it is recommended to use, for location angle θ *and* radius R variables in equations above, the following values, assuming valid the following transformation, for a section of height H :

$\alpha = \arcsin[H / (2 \cdot b)]$

$R' = \{ (H/2)^2 + [a \cdot \cos(\alpha)]^2 \}^{0.5}$ …or just R' equal to a or b , selecting one or other according to what is more conservative in formulas for internal loads calculation (eg. b for q_{max} and a for M_x)

$\theta' = \arcsin[H / (2 \cdot R')]$

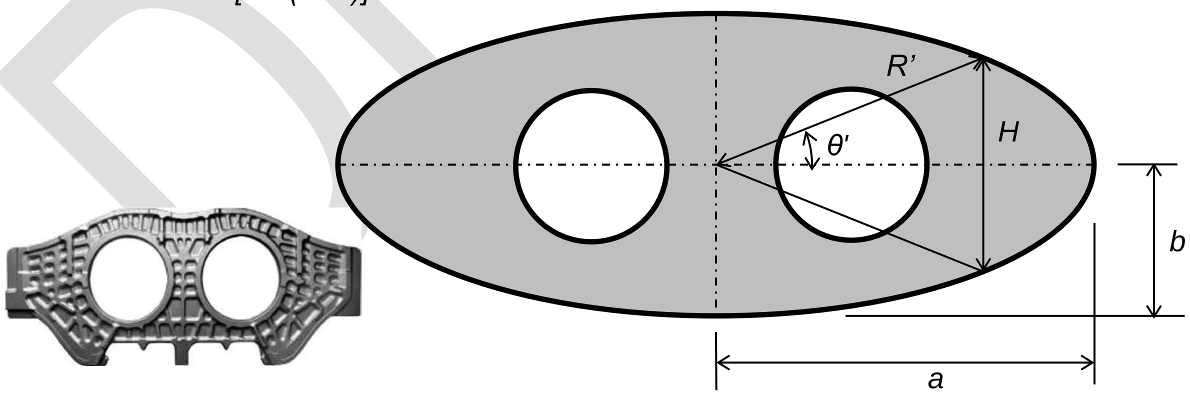


Figure 6. Picture of metal forging bulkhead for a twin-engine F-22 and ellipse transformation

- ✓ It is remarked that this methodology has still not been benchmarked or validated against reliable methods or tests. Thus, it should be used only for preliminary studies and presizing activities, and applying some extra penalizations to the already existing margins policies is strongly recommended.

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
A	mm^2	Area of the cross section
CG	-	As subscript, Centre of Gravity of the section
E	MPa	Young modulus
F	N	Force
fr	-	As subscript, refers to frame
F_{su}	MPa	Ultimate shear stress of a metallic material
F_{tu}	MPa	Ultimate tensile stress of a metallic material
I	mm^4	Bending moment of inertia of the cross section
KDF	-	Knock Down Factor
M	Nmm	Moment force
R	mm	Radius of the fuselage
sk	-	As subscript, refers to skin
σ	MPa	Axial/bending stress
τ	MPa	Shear stress
t	mm	Thickness
TBC		To Be Confirmed
X,Y,Z	-	Notation for the global reference axis system of the aircraft, where X is the longitudinal direction and Z is vertical positive upward

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[M.C.Y. Niu]	1998	AIRFRAME STRUCTURAL DESIGN
Ref. 2	[M.C.Y. Niu]	2nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 3	[W. C. Young & R. G. Budynas]	7th Ed.	ROARK'S FORMULAS FOR STRESS AND STRAIN

SOFTWARE

Program	***Version***	***Description***

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
0.1 19/04/2021	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Methods (TEPAP-TL4)	G. Baños 19/04/2021	F. Glock DD/MM/2021

Table 3. *Approval signatures table for last revision of this chapter*